

AI API and Operating Cost Report

Cost model, credit pricing, hosting, domain, database, and per-user economics

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Executive summary: A completed AI station should be priced at 5 credits, equal to PKR 25 or about USD 0.09. The estimated loaded cost is PKR 15-24, or USD 0.054-0.086, leaving a small but sustainable margin.

Key Recommendations

- Use a credit-based model for paid AI features instead of unlimited free AI use.
- Set 1 credit = PKR 5, approximately USD 0.018.
- Charge 3 credits for AI virtual patient mode and 2 credits for AI assessment.
- Charge 5 credits for a full AI station with assessment, equal to PKR 25 or approximately USD 0.09.
- Keep browser speech recognition free; include fallback transcription initially and monitor usage.
- Use Vercel Hobby only for MVP testing. Budget Vercel Pro for production.

Cost Snapshot

Item	Recommended Estimate	USD Equivalent
AI cost per complete station	PKR 10-15	USD 0.036-0.054
Loaded cost per complete station	PKR 15-24	USD 0.054-0.086
Recommended selling price	PKR 25	USD 0.09
Safer future selling price	PKR 30-40	USD 0.108-0.143
Normal active user AI cost/month	PKR 150-300	USD 0.54-1.08
Estimated total operating cost/user/month	PKR 250-450	USD 0.90-1.61

1. AI API Cost Estimate

The AI cost is driven primarily by the virtual patient conversation and the optional AI assessment. A normal station is assumed to include 20-25 student questions. The application is already cost-aware because it sends the current question, selected authored facts, and recent messages rather than the entire station script on every turn.

Usage Component	Estimated Usage Per Station
Virtual patient chat input	20,000-30,000 tokens
Virtual patient chat output	1,000-2,000 tokens
AI assessment input	3,000-5,000 tokens
AI assessment output	2,000-4,000 tokens
Total practical estimate	About 29,000 input + 5,000 output tokens

Estimated AI Cost Per Completed Station

Cost Level	PKR	USD	Notes
Low-cost model	PKR 1.5-3	USD 0.005-0.011	Cheapest possible model routing
Practical safe estimate	PKR 5-10	USD 0.018-0.036	Recommended for reporting
Higher-quality model	PKR 6-9	USD 0.022-0.032	Better grading quality

Voice and Transcription

The product should continue using the browser Web Speech API first because it does not create direct API cost. Groq Whisper Turbo can remain a fallback. At approximately USD 0.04 per audio hour, a fully recorded six-minute station would cost about USD 0.004, or about PKR 1.1. In normal usage, where only question snippets are recorded, the cost is expected to be lower.

2. Hosting, Domain, and Database Cost

Infrastructure Item	Monthly PKR	Monthly USD	Yearly PKR	Yearly USD
Vercel Hobby	PKR 0	USD 0	PKR 0	USD 0
Vercel Pro, 1 developer seat	PKR 5,580	USD 20	PKR 66,960	USD 240
Domain	-	-	PKR 5,000	USD 17.92
MongoDB MVP free tier	PKR 0	USD 0	PKR 0	USD 0
MongoDB small paid allowance	PKR 2,500-8,333	USD 8.96-29.87	PKR 30,000-100,000	USD 107.53-358.42

PKR 20,000 per year is enough for MVP deployment if free-tier hosting and database services are used. It is not enough for Vercel Pro production hosting. A safer production infrastructure budget before AI usage is PKR 100,000-170,000 per year.

3. Credit-Based Monetization Model

A credit model gives the platform control over API spend while making paid AI features easy for students to understand. Credits should only be consumed when the user uses paid AI features, such as virtual patient simulation or AI assessment.

Feature	Credits	PKR Value	USD Value
AI virtual patient station only	3 credits	PKR 15	USD 0.054
AI assessment only	2 credits	PKR 10	USD 0.036
Full AI station + assessment	5 credits	PKR 25	USD 0.09
Voice fallback	Included initially	PKR 0	USD 0

Credit Package Pricing

Package	Price PKR	Price USD	Credits	Full AI Stations	Effective Price/Station
Starter	PKR 999	USD 3.58	220	44	PKR 22.70 / USD 0.081
Standard	PKR 1,999	USD 7.16	480	96	PKR 20.80 / USD 0.075
Pro	PKR 2,999	USD 10.0	640	146	PKR 20.80 / USD 0.075

Profitability Per Full AI Station

Selling Price	Loaded Cost	Estimated Gross Profit
PKR 25 / USD 0.09	PKR 15-24 / USD 0.054-0.086	PKR 1-10 / USD 0.004-0.036
PKR 30 / USD 0.108	PKR 15-24 / USD 0.054-0.086	PKR 6-15 / USD 0.022-0.054
PKR 40 / USD 0.143	PKR 15-24 / USD 0.054-0.086	PKR 16-25 / USD 0.057-0.090

The recommended starting price is PKR 25 for a full AI station. This is viable if model routing remains optimized. For stronger margin and future price stability, the platform should be prepared to move full AI station pricing to PKR 30-40, or 6-8 credits.

4. User-Level and Yearly Operating Cost

Normal active user assumption: 30 full AI stations per month, 5 credits per station, 150 credits per month.

Metric	PKR	USD
User pays at 5 credits/station	PKR 750/month	USD 2.69/month
Estimated actual AI cost	PKR 300-450/month	USD 1.08-1.61/month
Loaded cost estimate	PKR 450-720/month	USD 1.61-2.58/month
Estimated gross profit	PKR 30-300/month	USD 0.11-1.08/month

Yearly Scenarios

Scenario	Included Assumptions	Estimated Yearly Cost
MVP / early testing	Vercel Hobby, MongoDB free tier, domain, 100 normal users	PKR 185,000-365,000 / USD 663-1,308
Small production	Vercel Pro, domain, MongoDB paid allowance, 100 normal users	PKR 282,000-532,000 / USD 1,011-1,907
Production infrastructure before AI	Vercel Pro, domain, MongoDB small paid allowance	PKR 101,960-171,960 / USD 365-616

Recommended Final Pricing Policy

Pricing policy: Set 1 credit at PKR 5, approximately USD 0.018. Charge 5 credits for a full AI station with assessment, equal to PKR 25 or USD 0.09. Review pricing once real production usage data is available.

- Use PKR 5-10 as the report estimate for AI cost per completed station.
- Use PKR 15-24 as the loaded internal cost per full station after infrastructure and buffer.
- Use PKR 25 as the minimum launch price for a complete AI station.
- Use PKR 30-40 as the safer future price range if usage increases or higher-quality models are used.
- Track actual input/output tokens, retries, speech fallback minutes, and assessment output length from day one.

Conclusion

The AI component is financially viable if access is controlled through credits. The proposed model keeps entry pricing affordable for students while protecting the platform from unlimited AI usage. At the recommended launch price of PKR 25 per full AI station, the platform can cover API costs and a share of infrastructure costs. For long-term sustainability and profit, larger credit bundles and a future full-station price of PKR 30-40 should be considered.